

Technology Stack

Date	30 September 2026
Team ID	SWTID-2026-8336
Project Name	EduGenie — Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, JavaScript, HTML5, CSS3, React Router, Axios	React.js provides a component-based and responsive user interface. React Router supports page navigation, while Axios connects the frontend with the backend REST API. HTML5, CSS3 and JavaScript support the overall web interface

2	Backend / Server-Side	Node.js, Express.js	Node.js provides the server-side runtime and Express.js is used to build REST APIs and handle authentication, learning requests, quiz operations and communication between the frontend, database and AI service.
3	Database / Data Storage	MongoDB Atlas, Mongoose	MongoDB Atlas stores user and learning-related data, while Mongoose provides structured interaction between the Node.js backend and MongoDB.
4	Cloud / Hosting / Deployment	Local Development	The current EduGenie application is developed and tested locally. It has not been deployed to a public cloud hosting platform
5	Version Control & CI/CD	Git, GitHub	Git and GitHub are used for source-code version control, project backup and collaborative development.

6	Third-Party APIs / Other Tools	Google Gemini API	Google Gemini API provides the AI-powered learning functionality of EduGenie, including educational assistance and AI-generated learning responses.s
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